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# Seeing the Process Through the Worker's Eyes: Event Logs from Egocentric Factory Videos

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**Abstract**

Much of manual factory work remains invisible to information systems: moving materials, preparing tools, checking surfaces, and repairing defects can leave no event record. Egocentric video brings this work into view, but turning what a camera sees into a process log requires decisions about actions, names, and time. We present a construction method and linked data resource that make these decisions inspectable. Pretrained models narrate recordings, build shared factory vocabularies from worker summaries, and convert the transcripts into labeled interval events. The resulting logs support exploration of work composition, local sequences, and differences between recorded worker profiles. Worked examples reveal how the representation shapes the findings: recorded transport can disappear from an aggregated transport category, and a narrated episode can acquire an additional event boundary during extraction. Structural checks and an author audit assess the records and their consistency with generated transcripts, while accuracy against the source videos remains an open question. By preserving the intermediate descriptions alongside the logs, the resource lets researchers trace measurements back to their textual basis and investigate alternative representations of manual work.

**Keywords:** process mining; inferred event logs; egocentric video; measurement; data quality; manufacturing; foundation models

## 1. Introduction

Process mining uses timestamped event records to study operational processes [1]. Many manual factory tasks leave no such records. An information system may record an order or machine checkpoint while missing how a worker loads parts, moves material, checks a component, or switches tools. Video can make some of this work observable, but it does not directly provide the case identifiers, labels, and event boundaries that process mining needs [2].

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**Figure 1.** Example frames from Build AI's Egocentric-10K resource [3]. They illustrate the camera viewpoint, rather than the accuracy of the derived event labels.

Egocentric video records work from the camera wearer's perspective. The public Egocentric-10K resource provides factory recordings of this kind [3]. Figure 1 illustrates the viewpoint. Turning recordings into event logs requires decisions about what constitutes an event, which names to use, and how to interpret time. These are measurement decisions as well as extraction tasks. A well-formed log can still misrepresent the work if a behavior is omitted, a boundary is estimated poorly, or an analytical category hides a recorded action.

Video-to-log construction already includes language-model-based methods. Vid2log uses a pretrained vision-language model to interpret video and construct event data [4]. First-person workflow extraction also predates this study [5,6]. The question here is how to build a shared event representation from the available factory recordings and explain its consequences for process analysis. In particular, naming an activity, assigning an episode to that name, and mapping the label into a measurement category are separate decisions. A shared vocabulary makes within-factory comparison possible, but does not by itself make the resulting measurements accurate or comparable across tasks.

We study a five-stage pipeline that produces timestamped transcripts, worker summaries, factory reports and vocabularies, per-video events, and merged logs. It induces a vocabulary across the recorded workers in each factory before assigning event labels from their transcripts. Its application covers six factories, 59 worker identifiers, and 322 recordings. The resulting 10,698 events are inferred operational episodes. They do not provide a complete census of production cycles or manufactured items.

The paper makes three contributions:

- **A construction design.** We specify how descriptive narration, aggregation across workers, factory vocabulary induction, and vocabulary-constrained extraction produce interval events. The foundation-model parameters remain fixed; the representation adapts to the available factory corpus. Formal definitions and implementation checks make the event and case choices explicit.
- **A reusable resource.** We link narrations, worker summaries, factory vocabularies, inferred intervals, consolidated logs, and analysis artifacts. This collection supports

**Table 1.** Research questions, evidence, and limits.

| Question and section | Supporting artifacts | Finding | Main limit |
|---|---|---|---|
| **RQ1**, Sections 4 and 7.1 | Scripts, prompts, reports, all per-video CSVs | Fields, ordering, and vocabulary membership can be checked | Valid structure does not imply correct observation |
| **RQ2**, Section 7.2 | Transcripts, logs, 84 archived audit judgments | Partial temporal and semantic consistency | No independent video reference; timestamp anomalies |
| **RQ3**, Section 7.4 | Frozen events, process-category map, reports | Composition, sequence, and worker-profile descriptions | Unequal windows, task mixes, and category meanings |
| **RQ4**, Section 8 | Worked examples, merge code, metric definitions | Aggregation can hide behavior; missing time has two-sided effects | Conceptual deductions do not estimate unknown errors |

inspection of the construction steps and reuse of the intermediate text. It is not an independently annotated ground-truth benchmark.

- **Findings about representation and measurement.** Existing examples show how event boundaries, case construction, and category maps affect duration and sequence descriptions. In factory **003**, transport is recorded but disappears from the transport category under the chosen process map. We separate these findings from transcript consistency, visual accuracy, and operational claims that require further evidence.

The assessment combines checks of the released records, archived author judgments against transcripts, and descriptive process analyses. Appendix A of the Additional Material summarizes the evidence supporting each type of conclusion.

Four research questions organize the paper. **RQ1:** How are the inferred logs constructed, and which structural properties do the released records satisfy? **RQ2:** How well do event timing and meaning agree with the generated transcripts? **RQ3:** Which descriptive analyses do the logs support, and what limits their operational interpretation? **RQ4:** What general measurement lessons follow from the representation, temporal rules, and shared evidence? Table 1 links each question to its evidence.

## 2. Background and Related Work

### *2.1. Unstructured observations as process data*

König et al. [7] review how unstructured data enters process mining, including the construction of logs from text, images, and video. Diba et al. [8] distinguish extraction, correlation, and abstraction. These tasks require decisions about which observations become events, how events are grouped, and which concepts describe them. Andrews et al. [9] make data quality part of construction in RDB2Log, which helps analysts derive logs from relational data. Our setting adds a chain of inferred descriptions and labels before a log exists. We examine how choices along that chain affect the resulting measurements.

### *2.2. Video and first-person workflow understanding*

Petersen et al. [5] model and track manual workflows from first-person video to support augmented-reality instructions and feedback during task execution. Pham and Xiao [6] extract assembly steps and operations with semantic descriptions, without pre-labeled training data or pre-trained classifiers. These studies establish first-person workflow understanding as an antecedent. Their goals include following a demonstrated task and recovering its steps.

Our objective is retrospective process analysis across recorded workers. We construct event data to describe time composition, changes between operational categories, and variation in recorded work. A shared vocabulary lets these descriptions use the same names within a factory. It does not establish that workers performed equivalent tasks or

that a category transition was a physical interruption. Egocentric video provides access to the wearer's work; the analytical objective and event construction define its role here.

Video process mining also has several established designs. Knoch et al. [10] use hand trajectories and clustering at a camera-equipped assembly workstation. Kratsch et al. [2] develop a reference architecture, and Lepsien et al. [11] combine object tracking, action detection, and abstraction. SnapLog [12] uses pretrained visual representations, temporal segmentation, and a classifier adapted with labeled examples. Its evaluation includes TUM Kitchen and egocentric EPIC-KITCHENS recordings. Thus, camera placement and supervision must be compared for individual methods. The sensor-data review by Brzychczy et al. [13] places these efforts within the broader problem of converting observations into process events.

#### 2.2.1. From video descriptions to process events

Dense video captioning already combines temporal localization with natural-language descriptions [14]. Timestamped narration alone is therefore an established output. Process analysis requires further choices: reusable activity names across recordings, event granularity, selection of operationally relevant episodes, observational case identifiers, source links, and rules for interpreting durations and frequencies.

S1's instruction to merge stable repetition illustrates why these choices matter. Several repeated actions can become one narrated interval and later one event, so event frequency need not equal action frequency. Conversely, the existing foundry example in Section 7.3 turns one 31-second description into two events with shared factory labels and an inferred internal boundary. The added structure enables calculations, while changing what each row and duration represents.

### *2.3. Video-to-log and language-model-based construction*

Gavric et al.'s Vid2log [4] is a close predecessor. It uses LLaVA to infer a domain and relevant activities from selected frames, obtains textual event information, and combines video-derived data with conventional logs through temporal and semantic matching. Both methods use natural language as an interface between visual observations and process event data. Our design first narrates recordings, aggregates descriptions by worker, and induces shared factory label lists before assigning events from the original transcripts. This ordering makes the available factory corpus part of the representation used to analyze it. Vid2log studies log richness and an assembly use case; our assessment examines transcript consistency and the consequences of event and category choices for duration and sequence analyses. The comparison centers on how factory context is constructed and how the resulting event representation is assessed.

Kecht et al. [15] use pretrained natural language inference to assign supplied topics and activities to customer-service messages. They do not train a new language model, but use labeled messages to select classification thresholds and refine hypotheses. Message identifiers and timestamps are retained. Our method instead estimates episode boundaries from generated video descriptions and induces label lists from the factory material. Shirali et al. [16] also apply LLMs to physical observations: they abstract IoT readings into high-level events and integrate logs from multiple sensor sources. LLM use for physical-process data therefore predates this study. Work on process-mining assistants [17] and database-query generation [18] addresses neighboring tasks with different inputs and outputs.

Seeth et al. [19], an arXiv preprint, study fine-tuning for text-to-log generation. They pair generated reports with source traces, supply permitted activities and attributes, and evaluate both traces and discovered process models against source-log references. They report benefits from fine-tuning over zero-shot and few-shot prompting in that setting.

**Table 2.** Closest construction methods: purpose, input, vocabulary, adaptation, and output. "Not reported" identifies information not established by the inspected source, not an absent capability.

| Method | Purpose and input | Vocabulary origin | Task-specific adaptation | Output |
|---|---|---|---|---|
| Vid2log [4] | Construction and enrichment from video, with conventional logs | Domain and activity prompts | Pretrained LLaVA; domain prompts; parameter updates not reported | Combined event log |
| Kecht et al. [15] | Construction from customer-service conversations | Supplied topics and activities | Pretrained NLI; labeled threshold calibration and hypothesis selection | XES log |
| SnapLog [12] | Event discovery from video | Classes supported by labeled examples | Pretrained visual models; classifier adaptation | Events with labels or label distributions |
| Seeth et al. [19] | Construction from textual reports | Allowed activities from source logs | Pretrained LLMs; supervised fine-tuning compared with prompting | XES traces |
| This resource | Construction from egocentric factory video | Shared lists induced from worker summaries | Fixed pretrained models; factory-specific context and labels | Interval CSVs and linked text |

**Table 3.** Time, case relationships, and evaluation references for the same methods. A source timestamp, an estimated interval, and an artificial ordering carry different evidence.

| Method | Origin of temporal information | Origin of case/entity relationships | Evaluation reference |
|---|---|---|---|
| Vid2log | Frame positions and prompted temporal relations | Temporal/semantic matching and resource-based case inference | Assembly case study and log-richness assessment |
| Kecht et al. | Recorded message timestamps | Supplied message and conversation identifiers | Labeled messages for classification; process discovery demonstration |
| SnapLog | Segmented frame sequences | Video sequences as traces | Video datasets with action and boundary annotations |
| Seeth et al. | Timestamps extracted from generated reports | Report-to-trace pairing from source logs | Source traces and process-level comparisons |
| This resource | Estimated video-relative intervals; S4 may split spans; synthetic S5 timeline | Source factory/worker/video metadata; chosen video or worker grouping | Generated transcripts and author audit; computations on inferred logs |

Our models receive no task-specific parameter updates; adaptation takes place through staged context and a vocabulary induced from each factory corpus. A permitted label list is shared by both designs and is not itself the distinction. The studies differ in source material, reference data, and evaluation targets. Our assessment establishes what the reported procedure supports, not that fine-tuning is generally unnecessary. Numerical scores from these studies cannot be ranked as equivalent measures of extraction accuracy.

Tables 2 and 3 compare the closest transformations using the same method rows. Section 4.3 distinguishes model adaptation from human input in our study.

### *2.4. Event abstraction and semantic consistency*

The review by van Zelst et al. [20] explains how event abstraction changes process granularity and distinguishes methods by matters including supervision, input, and uncertainty. Here, three questions need separate treatment: where an episode begins and ends, which reusable activity classes exist, and which class is assigned to each episode. S1 mainly supplies candidate intervals, S2 and S3 construct the classes, and S4 assigns events and can revise boundaries. Section 4.1 maps these tasks to the implementation.

Inducing classes and assigning instances can fail in different ways. A useful concept can be missing from the vocabulary, or an available concept can be assigned incorrectly. A later category map can also hide a correctly recorded label. The activity and process fields are two label dimensions; the implementation does not establish a parent-child hierarchy. This distinction is necessary to explain factory **003**, where transport activities occur under process contexts that are mapped to other operational categories.

**Table 4.** Neighboring event-data tasks, distinguished by their starting points and intended results. A system can perform more than one task.

| Task | Starting point | Intended result |
|---|---|---|
| Construction | Observations or documents without a suitable log | An initial event representation |
| Enrichment | An existing log and additional information | Added or corrected events and attributes |
| Integration | Multiple sources describing related behavior | A combined representation with resolved correspondences |

### *2.5. Construction, enrichment, and integration*

Text2EL+ [21] enriches existing logs with events and attributes extracted from text, using semantic and contextual validation and expert guidance. Sturm et al. [22] instantiate parsing, abstraction, extraction, integration, and enrichment using CRM data, email metadata, and call transcripts. Their evaluation involves a sales process and a domain expert. Table 4 separates the tasks these designs can combine.

Our main task is construction. S1 to S4 derive an initial representation; S5 consolidates it using source filenames, factory and worker identifiers, and a stated ordering rule (Section 5.4). It does not resolve product correspondences between independently recorded systems. Videos, narrations, and reports are successive representations of the same originating observations. Agreement among them does not provide the independent corroboration that separately recorded enterprise or sensor sources could supply.

### *2.6. Quality, uncertainty, and relational representations*

Suriadi et al. [23] organize recurring event-log imperfections. Our contribution to this concern is an account of how narration, vocabulary construction, assignment, and consolidation can introduce or propagate them. Section 8.2 connects each transformation to an existing check or example and its measurement consequence.

Quality annotations and explicitly uncertain events serve different purposes. Goel et al. [24] retain data-quality information for use during analysis. Pegoraro et al. [25] represent uncertainty explicitly and compute conformance bounds. SnapLog can retain candidate-label distributions [12]. Our events instead have single-valued activity and process assignments. The audit records particular disagreements, and the five-second boundary tolerance is an evaluation rule. Neither provides calibrated event probabilities or bounds on total downstream error. Structured quality flags and explicit alternatives are possible extensions, with different evidence requirements.

Esser and Fahland [26] represent events and entities across multiple dimensions in graph databases. This complements the case-choice concerns of object-centric process mining [27]. Our worker and video relationships support alternative observational views. Product identities and cross-worker product trajectories require additional evidence; a change in storage format cannot establish them. Similarly, the range of industrial process-mining goals [28] makes it necessary to distinguish descriptions of recorded work from conclusions about production performance.

The question addressed here is how a shared representation, built from the available factory recordings before event assignment, supports retrospective analysis and shapes its results. **RQ1** examines that construction and its structural properties. **RQ2** checks the transcript-to-event transformation. **RQ3** studies the supported calculations, and **RQ4** explains their dependence on vocabulary, boundaries, case rules, and analytical categories. Together, these questions connect the construction design, artifact release, and measurement findings to their methodological antecedents.

**Table 5.** Notation and intended meaning.

| Symbol | Meaning |
|---|---|
| $f, w, v$ | Factory, worker, and video identifiers; identifiers do not certify roles or task equivalence |
| $D_v, T_v$ | Source duration and generated transcript; the former is unavailable in the released subset |
| $A_f, P_f$ | Two separate permitted label lists for factory $f$ |
| $t_s, t_e, d_e$ | Estimated video-relative start, end, and duration $d_e = t_e - t_s$, in seconds |
| $\kappa(e)$ | Selected case identifier, such as the source video |
| $g_f(p)$ | Analyst's mapping from process label to operational category |
| $C_v, S_v$ | Interval-union event duration and endpoint span used in coverage diagnostics |

## 3. Problem Definition

The starting point is the factory video in the Egocentric-10K dataset. A camera worn by a worker records visible actions involving tools, materials, and work surfaces. These recordings contain evidence of work, but they do not directly identify the interval events needed for process mining. We therefore first use a vision-language model to produce a timestamped natural-language transcript of each video. We then use large language models (LLMs) to organize the descriptions into shared factory vocabularies and convert the transcripts into interval events. Each event should state what action occurred, its broader process context, and its estimated start and end within the recording.

Let $F$ be the set of factories, $W_f$ the worker identifiers associated with factory $f$, and $V_w$ the recordings associated with worker $w$. Each video $v$ has an underlying duration $D_v$, which need not be available in the derived archive. Its generated transcript $T_v$ is an ordered collection of entries $(s_i, e_i, x_i)$, where $x_i$ describes the interval from $s_i$ to $e_i$. The descriptions and endpoints are model outputs, so the transcript is an intermediate representation of the video rather than an independent annotation.

For each factory, the pipeline induces an activity vocabulary $A_f$ and a process vocabulary $P_f$. An interval event is

$$e = (v, w, f, t_s, t_e, a, p), \qquad a \in A_f, \quad p \in P_f. \tag{1}$$

The tuple makes provenance explicit by linking an event to its source recording, worker, and factory. The estimated endpoints $t_s, t_e$ are relative to the video. The LLM interprets $T_v$ using $A_f$, $P_f$, and the worker context to produce the event set $E_v$; the complete log contains $E = \bigcup_v E_v$. A case function $\kappa : E \to K$ then groups events into traces for a chosen analysis. The construction problem is thus to transform videos into transcripts and transcripts into labeled intervals, while retaining enough source information to inspect the resulting log. Vocabulary induction uses descriptions across the available factory recordings, so event extraction depends on both the local transcript and its factory context.

Table 5 summarizes the notation. An event represents an operational episode. One episode may summarize many repeated actions, and several events may divide a single narrated interval. It need not be one item, one production cycle, or one complete process instance. Desirable properties include faithful descriptions, meaningful boundaries, appropriate labels, and sufficient coverage. The implementation enforces only a subset: parseable offsets, end not before start, required values, and membership in the separate vocabularies. Section 4.3 distinguishes these checks from intended properties.

### 3.1. Activity, process context, and analytical category

An *activity* answers what action is performed. A *process context* identifies the broader operation to which the extractor assigns it. An *operational category* answers how an analyst

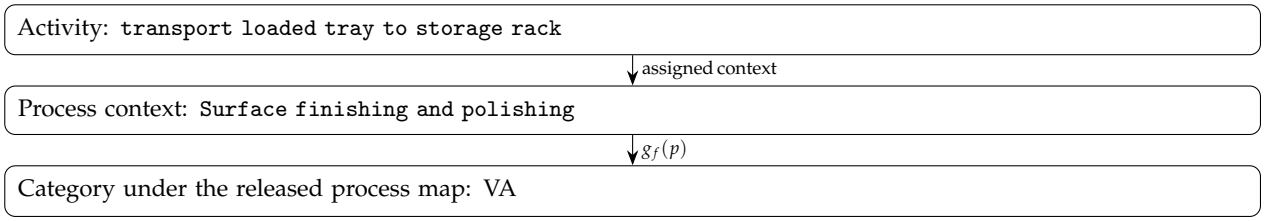



**Figure 2.** A transport action can contribute to VA time under a process-based category rule. Source: factory **003**, worker 001, recording 00001, event row 2, from 00:03:52 to 00:04:00 (8 seconds). The action is recorded; the category aggregation hides it.

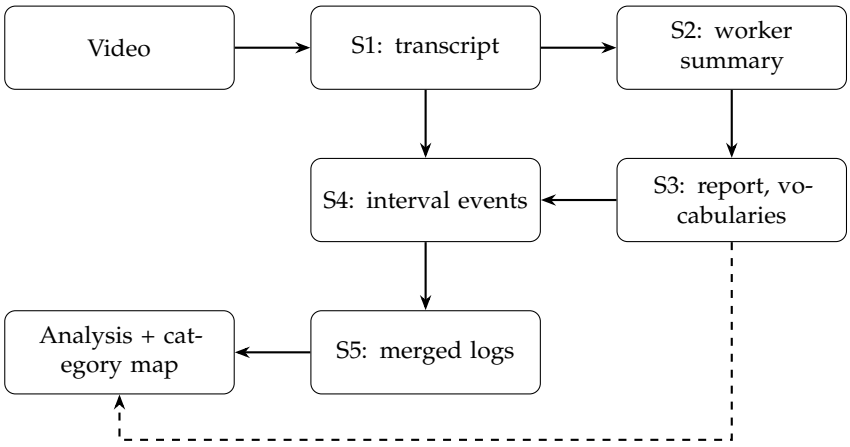



**Figure 3.** Evidence dependencies in the implemented pipeline. S1 to S3 use Gemini requests; S4 uses a text-only OpenAI request. The analysis map is an author-defined interpretation. An error in the transcript can affect both the narrative report and the extracted log.

groups the action or context for a particular question. These concepts are separate. The implementation checks $a \in A_f$ and $p \in P_f$ independently. It does not enforce a compatibility relation $H_f \subseteq A_f \times P_f$ or a process-to-activity hierarchy.

The current time analysis uses $g_f(p)$, not an activity-based mapping. Thus, transport within surface finishing can receive the category assigned to finishing. Figure 2 gives an exact released example. An explicit compatibility relation or a category rule using $(a, p)$ could improve future analyses, but neither was enforced in this extraction. An activity-based category rule would define a different measurement of the recorded work.

## 4. Pipeline and Design Rationale

The released scripts implement five stages. S1 uses video and a transcription prompt. S2 combines a worker's transcripts. S3 combines a factory's worker summaries. S4 uses the original transcript again, together with the factory vocabularies and the compact worker entry from the S3 report. S5 merges the resulting CSVs. Figure 3 shows these dependencies. Reports and logs share evidence, so their agreement is not independent validation.

Adaptation takes place in the context and vocabulary, while the foundation-model parameters remain fixed. Each available factory corpus contributes to its own label lists through S2 and S3, and S4 then uses those lists to label recordings from that corpus. The six-factory assessment therefore applies the same construction procedure to six separately induced representations. It is a retrospective use of the available material, not an evaluation of a fixed vocabulary on a previously unseen factory. Prompts also supply analyst choices about operational relevance and granularity.

**Table 6.** Conceptual tasks and their roles in the five-stage construction. The stages interact; this is not a claim that each task has an isolated error rate.

| Task | Main stage | Decision made |
|---|---|---|
| Temporal segmentation | S1 | Describe candidate episodes with estimated boundaries; merge stable repetition |
| Activity-class construction | S2 and S3 | Aggregate worker descriptions and induce reusable factory activity and process lists |
| Event-instance assignment | S4 | Select permitted labels for episodes; retain or split transcript intervals and estimate internal boundaries |

**Table 7.** Stage specification from scripts 01 to 05. A nonempty text output is not a validation of its factual content.

| Stage; unit | Inputs and context | Output | Failure and reuse behavior |
|---|---|---|---|
| S1; video | Uploaded video; transcription prompt | One chronological text file | Keeps existing files; retries transient failures; rejects an empty response; attempts deletion of uploads |
| S2; worker | All saved transcripts in one request; filenames omitted from the bundle | Worker report | Missing or empty input raises an error; empty response rejected; keeps existing output unless overwrite is set |
| S3; factory | All worker summaries in one request, with worker identifiers | Factory report with two label lists and compact worker entries | Same text and reuse checks as S2; malformed section structure may fail later in S4 |
| S4; transcript | Transcript; two label lists; compact S3 worker entry, or a missing-context marker | Per-video CSV | Missing lists or invalid events raise errors; an empty event list is valid; existing CSV kept unless overwrite is set |
| S5; worker, then factory | CSVs in sorted filename order | Worker and factory CSVs | Checks timestamp columns and consistent headers; empty files add no time; keeps existing output if its schema matches |

```
for each selected factory f:
    for each available video v:
        T[v] = reuse existing transcript or S1(video[v], prompt1)
    for each worker w:
        B = concatenate all T[v] for w, without filenames
        R[w] = reuse existing summary or S2(B, prompt2)
    F[f] = reuse existing report or S3(all R[w] for f, prompt3)
    A, P, worker_context = parse the required sections of F[f]
    for each transcript T[v]:
        context = T[v] + A + P + worker_context[w]
        E[v] = reuse CSV or normalize(S4(context, prompt4))
    for each worker w:
        offset = 0
        for each CSV v in sorted filename order:
            append E[v] shifted by offset; keep local offsets
            if E[v] is nonempty: offset += max end(E[v])
    concatenate worker logs into the factory log
```

Listing 1: Algorithm 1: implemented dependency order. API retries surround requests; this pseudocode does not add windowing or new validation.

### *4.1. Vocabulary construction as event abstraction*

Event abstraction concerns the level at which observations represent process behavior [20]. Table 6 separates three decisions in our implementation. Constructing a reusable class and assigning an event to it are different tasks: a missing concept and a mistaken assignment require different checks. The process list supplies a second label dimension, with no enforced parent-child relation to the activity list.

### *4.2. Execution units and failure behavior*

Table 7 states the input, output, and failure behavior for each execution unit. The algorithm below gives their dependency order.

S1 asks for a new timestamped entry when an action, object, location, or interaction changes. It asks for repeated stable work to be summarized once and for uncertainty to be

**Table 8.** Four levels of evidence for pipeline properties.

| Level | Example | What can be concluded |
|---|---|---|
| Prompt instruction | Mark uncertainty; include handling and waiting when visible; split meaningful changes | The behavior is requested. Compliance and completeness require separate assessment. |
| Software constraint | Parse event fields; canonicalize permitted labels; reject negative intervals; sort rows | Accepted outputs satisfy these checks. Semantic correctness and accurate time remain open. |
| Observed artifact property | All 10,698 rows have required values and permitted labels; 35 adjacent overlaps are detected | These statements hold for this release. They are not promises about future runs. |
| Untested design expectation | Summaries help vocabulary consistency; closed labels help interpretation | These are rationales. No ablation or comparative benefit estimate is available. |

**Table 9.** Supervision and human input in this study. Unlike labeled threshold calibration in Kecht et al. [15] or parameter adaptation in SnapLog and Seeth et al. [12,19], generation here adapts through prompts and induced context.

| Dimension | Role in the present procedure |
|---|---|
| Model pretraining | General-purpose pretrained models provide visual and language capabilities |
| Task-specific parameter updates | None in S1 to S4; no model is trained or fine-tuned on these factory recordings |
| Labeled examples | No labeled event examples for training, prompting, or calibration; S1 includes an illustrative narration-format example |
| Vocabulary origin | S3 induces two lists from the available worker summaries; S4 checks list membership |
| Domain and analyst input | Authored prompts define relevance and segmentation; analysts choose case projections and the operational-category map |
| Human assessment | One author's archived event and transcript-entry judgments assess outputs; they are not generation inputs |

stated briefly. These are instructions to the model. S1 does not check whether each visible change was narrated or whether the timestamps match the video.

S2 and S3 compress large text bundles into summaries. This makes factory-wide vocabulary induction possible, but uncommon actions can be lost. The implementation does not automatically split long requests into windows, reconcile separate summaries, or check coverage against all transcript spans. S4 partly bypasses this compression by reading each original transcript, although it must still use the induced vocabularies.

S4 requests a strict JSON schema with estimated start and end times, activity, and process. Its local parser can also accept a list, strip code fences, recognize fallback time keys, and parse two-component offsets or numeric seconds. It normalizes label case and surrounding quotes, checks membership, rejects end-before-start intervals, and sorts rows by start, end, activity, and process. It allows zero-duration events and overlaps. It does not check video bounds, full temporal coverage, the truth of an activity, or compatibility between the activity and process. The archive contains CSV outputs, not the original S4 JSON responses.

S1 to S3 default to `gemini-3.1-pro-preview`; S4 defaults to `gpt-5.4`, requests high reasoning effort, and permits up to 100 concurrent extraction tasks. These are recoverable code settings. Per-request execution records are unavailable. No temperature, token limit, seed, cost, or runtime is inferred from missing records. Appendix B of the Additional Material specifies the distinction between code defaults and historical execution evidence.

### *4.3. Requests, guarantees, observations, and expectations*

Table 8 distinguishes requested, enforced, observed, and untested properties. "Automated extraction" means that generation uses no task-specific fine-tuning or labeled event examples. It does not mean that the complete study is free of human judgment. Table 9 identifies the different inputs. "Constrained" refers to the separate permitted label lists. "Validated" is used only with a named check, never as a general claim of visual accuracy.

**Table 10.** Design rationale and conceptual alternatives. No alternative in the last column was experimentally compared here.

| Choice | Intended benefit | Limitation | Possible alternative |
|---|---|---|---|
| Video-to-text intermediate | Readable evidence; reuse for different analyses | Visual errors become shared textual evidence | Direct video-to-events, with stronger visual provenance |
| Worker summaries | Compact factory-level context | Rare behavior and recording identity may be lost | Window summaries with explicit source references |
| Factory vocabularies | Concrete local labels across workers | Different factories use different abstractions | Global concepts with local extensions |
| Closed labels | Prevent out-of-list output | Closest allowed label may still be wrong | Open labels followed by review and reconciliation |
| Change-point narration | Compact description of stable behavior | Boundary detail and repeated cycles can disappear | Finer segments with explicit repetition counts |
| Interval events | Preserve approximate episode duration | Row counts do not equal production volume | Object-linked cycle records, when supported |

### *4.4. Design choices and tradeoffs*

Table 10 summarizes the design choices and alternatives. Repeated cycles deserve particular care. If ten cycles become one narrated interval and one event, the event count decreases and the directly-follows relation loses nine possible repetitions. A longer event can represent more repetitions, slower work, or more aggressive compression. The logs alone do not distinguish these explanations. The text interface makes this decision inspectable, but does not recover details omitted during transcription.

## 5. Dataset and Released Artifacts

### *5.1. An inferred resource for process analysis*

Assembly101 [29] records toy assembly and disassembly with simultaneous static and egocentric cameras. Its coarse and fine action annotations support benchmarks for recognition, anticipation, and temporal segmentation. Our source setting is recorded manual work in six factories, and the released labels and boundaries are inferred by the pipeline. The distinction is the purpose and provenance of the resource: linked narrations, summaries, vocabularies, events, logs, and analysis artifacts make the construction and interpretation of process measurements inspectable.

Researchers can use the preserved narrations to study alternative event representations, inspect which descriptions support a label, or compare how activity and process categories affect an aggregate. Such comparisons would examine representation choices unless independently checked against video. The present release includes structural checks and an author audit against generated transcripts, but no independent video annotations. Its inferred labels are not an independently annotated ground-truth benchmark for action recognition or boundary detection.

### *5.2. Selection and provenance*

The archived subset contains outputs for factories **001** to **006**, with 59 worker identifiers and 322 paired transcript and event files. Factory and worker identifiers are inherited from the source organization. Domain descriptions in this paper summarize generated factory reports; they are not independently verified factory metadata. The archive does not record the selection rationale, the original source revision, or a complete inventory of the recordings available at extraction time. We therefore describe an available processed subset, not a random or representative sample of factories.

The current provider card reports a median clip length of 180 seconds [3], whereas many archived transcript endpoints lie near 20 minutes. These describe different quantities and possibly different snapshots or subsets. The card documents a per-video `duration_sec`

**Table 11.** Artifact inventory. Counts come from the frozen repository outputs. Worker and factory merged logs repeat the same source events.

| Layer | Files | Content and provenance |
|---|---|---|
| Transcripts | 322 | One text per video; recording identity in path |
| Worker summaries | 59 | One text per worker; combines transcripts |
| Factory reports | 6 | Narrative, worker entries, activity and process lists |
| Per-video logs | 322 | 10,698 interval-event rows; six stored columns |
| Merged worker logs | 59 | Same 10,698 events with source file, worker case, and local offsets |
| Merged factory logs | 6 | Same 10,698 events concatenated across workers |
| Audit record | 1 | 42 event judgments and 42 transcript-entry judgments |
| Prompts; stage scripts | 4; 5 | Extraction instructions and implementation |

**Table 12.** Transcript and event endpoints for the seven literal transcript spans over one hour. The final column applies the original time-analysis rule to event endpoints. IDs abbreviate factory/worker/video. Source durations are unavailable for all rows.

| ID | Transcript endpoint | Event endpoint | Time-analysis decision |
|---|---|---|---|
| **001**/001/00004 | 20:00:00 | 20:00:00 | Exclude |
| **001**/010/00001 | 01:20:00 | 01:20:00 | Exclude |
| **002**/002/00008 | 20:00:00 | 00:20:00 | Retain |
| **002**/010/00001 | 20:00:00 | 00:20:00 | Retain |
| **005**/002/00000 | 13:52:00 | 00:13:52 | Retain |
| **006**/001/00001 | 20:06:00 | 00:20:06 | Retain |
| **006**/004/00000 | 19:59:00 | 00:19:59 | Retain |

field inside source archives, but those metadata files are not included here. The pipeline scans video files and does not preserve that field. The selected recordings' actual durations therefore cannot be established from the derived archive. The inspected public dataset snapshot does not identify the historical source version used for extraction.

The six per-video columns are `start_timestamp`, `end_timestamp`, `activity`, `process`, `factory`, and `worker`. S5 adds `source_file`, `case_id`, video-relative start and end offsets, and duration. Table 11 and the hashed artifact manifest distinguish file counts from unique event counts. The released dataset documentation follows the documentation concerns of Datasheets for Datasets [30]: composition, collection, uses, and limitations.

There are 13,304 entries recognized by the original three-component timestamp parser. Recognizing the additional two-component and mixed-format entries gives 13,462 entries, without rescaling any timestamp. The original endpoint-span calculation totals 786,412 seconds (218.4 hours). This is a sum of parsed spans, not verified footage duration. Across all logs, interval unions cover 414,886 seconds and summed event durations total 421,283 seconds. Their difference, 6,397 seconds, reflects overlaps. The retained time-analysis set contains 10,647 events in 320 logs, with 349,659 seconds of summed event duration (97.1 hours). These quantities must not be interchanged.

### *5.3. Temporal data quality and inclusion rules*

Table 12 identifies all seven transcripts whose literal parsed endpoint exceeds one hour. An endpoint of `20:06:00` paired with an event endpoint of `00:20:06` is consistent with a format error, but does not prove which timestamp is correct. Some files switch from `HH:MM:SS` to `MM:SS`. The analysis retains the recorded timestamps.

The original value-analysis code excludes a log only when its maximum event endpoint exceeds 3,600 seconds. It excludes two logs, both in factory **001**. This rule retains five logs associated with timestamp-anomalous transcripts. Retention is a calculation rule, not certification of their timestamps.

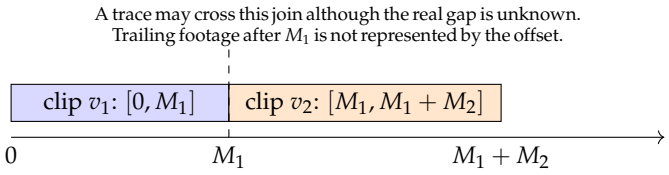



**Figure 4.** Schematic of the implemented merge, not measured source time. Separate video cases preserve the clip boundary. A worker case creates a directly-follows relation across it.

**Table 13.** Case constructions supported by the attributes. Process episodes require an additional grouping step and are not stored as case IDs.

| Case rule | Supported description | Interpretation risk |
|---|---|---|
| Video as case, $\kappa(e) = v$ | Local activity order within an observation window; used for discovery | Window may start or end midway through a production process |
| Worker as case, $\kappa(e) = (f, w)$ | Sequence of retained clips; stored S5 case ID | Filename joins can create artificial transitions; calendar duration is unknown |
| Contiguous process episode | Consecutive events with the same process within a chosen clip, with a stated gap rule | Inferred context boundaries do not identify items or completed instances |

Four other transcripts contain two-component endpoints. Their identifiers are **001**/004/00001, **002**/005/00001, **002**/009/00003, and **003**/003/00003. Three change format within the file; **002**/005/00001 uses `MM:SS` throughout. The original parser misses 158 entries across these files. The inclusion manifest records every video's formats, format-transition positions, endpoints, row count, union duration, summed duration, and decision. A separate interval manifest preserves the original endpoint strings. The low coverage ratios in Section 7.2 therefore diagnose inconsistent representations; they do not establish long-video truncation.

### *5.4. Synthetic time and case selection*

S4 writes offsets as dates relative to 2000-01-01. This date exists for software compatibility. The archive provides no verified calendar times or cross-worker synchronization. S5 processes clips in filename order, which is not proof of their real chronological order. For a worker's ordered clips $v_1, \ldots, v_m$, define $M_j = \max_{e \in E_{v_j}} t_e$, with $M_j = 0$ for an empty log. S5 uses

$$O_1 = 0, \qquad O_{j+1} = O_j + M_j, \qquad (t'_s, t'_e) = (O_j + t_s, O_j + t_e). \tag{2}$$

The next clip starts at the largest extracted endpoint, not at the full source duration. Unrepresented trailing footage and empty logs add no elapsed time. Within-clip gaps before that endpoint remain on the synthetic axis, but gaps between recording sessions are unknown. Each worker has a separate synthetic origin; equal synthetic dates do not establish simultaneous work.

Figure 4 illustrates the synthetic clip join, and Table 13 compares the case rules supported by the released attributes.

## 6. Assessment Methodology

The assessment uses frozen texts, reports, CSVs, and archived author judgments. The descriptive workflow runs offline with Python's standard library. Activity discovery uses PM4Py [31] with the archived miner and noise-threshold settings. All calculations use the released records as their input.

**Table 14.** Evaluation questions, required references, and evidence in this study. Successful computation and correct operational interpretation are separate claims.

| Question | Reference needed | Present evidence and scope |
|---|---|---|
| Are visual actions correct? | Independent annotations of the video | Not measured; generated transcripts cannot detect their own visual errors |
| Are labels consistent with text? | A reference interpretation of the text | Existing author judgments for 42 events; label membership checked for all rows |
| Are boundaries accurate? | Independently defined action boundaries | Agreement with generated transcript endpoints only; the five-second tolerance is a matching rule |
| Can the log support process-mining computations? | Defined case rules, measures, and completed calculations | Discovery and descriptive outputs demonstrate computation on the inferred representation |
| Are operational conclusions correct? | Valid measurements and evidence relevant to the conclusion | Traceable category descriptions and hypotheses; no independent production outcomes or intervention evidence |

### *6.1. Evaluation questions and reference data*

Table 14 distinguishes the objects of evaluation. Assembly101 provides video action annotations [29]; Seeth et al. compare text-derived logs with source traces and process-level references [19]. Our transcript checks assess an intermediate transformation, while our downstream calculations demonstrate uses of its output. These reference structures support different conclusions, so their accuracy, fitness, and precision values should not be ranked as if they measured the same property.

### *6.2. Metrics, denominators, and aggregation*

For each video, let $E_v$ be its events and $B_v$ the set of transcript endpoints recognized by the original parser. Let $T_v^{\max}$ be the maximum such endpoint and $M_v$ the maximum event endpoint. The original coverage diagnostic uses

$$C_v = \left| \bigcup_{e \in E_v} [t_s, t_e) \right|, \qquad S_v = \max(T_v^{\max}, M_v), \qquad \text{coverage}_f = \frac{\sum_{v \in f} C_v}{\sum_{v \in f} S_v}. \tag{3}$$

The denominator begins at zero. This is an endpoint-span ratio, weighted by span, not a mean of clip ratios and not footage coverage. When no transcript endpoint is recognized, the event endpoint supplies the denominator. The corrected parser's results are released as a separate diagnostic; the original table is retained and labeled.

A boundary matches if its distance from *any* recognized transcript endpoint is at most five seconds. Both ends of every event enter the denominator, including events in files with no recognized transcript entries. Multiple event boundaries can match the same transcript boundary. A match need not identify the correct action or justify an internal split.

**Table 15.** Metric definitions. $E_f^*$ is the retained set under $M_v \leq 3600$. $E_w^*$ is its worker subset, $H_w = \sum_{e \in E_w^*} d_e / 3600$, and $R_w^{VA}$ is the worker's list of VA run durations.

| Quantity | Definition | Unit, aggregation, exclusions |
|---|---|---|
| Label conformance | $\#\{e : a \in A_f,\ p \in P_f\} / \#E_f$ | Fraction of all event rows |
| Vocabulary use | Distinct assigned labels divided by permitted labels | Separate activity and process lists, per factory |
| Coverage diagnostic | $\sum C_v / \sum S_v$ | Fraction of parsed endpoint span; all logs |
| Boundary agreement | $\sum_{e,b \in \{t_s,t_e\}} \mathbf{1}[\min_{t \in B_v} \lvert b - t \rvert \leq 5] / (2\#E)$ | Fraction of boundaries; all logs; original parser |
| Category share | $q_{fc} = \sum_{e \in E_f^*} d_e \mathbf{1}[g_f(p_e) = c] / \sum_{e \in E_f^*} d_e$ | Fraction of summed event duration; overlap counted in each row |
| Worker VA share | Same ratio restricted to $E_w^*$ and category VA | Eligible workers require $H_w \geq 0.5$; factory median weights workers equally |
| VA transition rate | $I_w / H_w$, where $I_w$ counts VA-to-nonVA transitions in concatenated run lists | Transitions per summed event-hour; factory median |
| Mean VA run | $\overline{R}_w = \sum R_w^{VA} / \#R_w^{VA}$ | Minutes; table reports median across eligible workers of these means |
| MH/TR count and rate | Number of retained MH or TR rows; count divided by summed event-hours | Rows, not merged episodes; duration-weighted factory denominator |
| Dominant process | $\arg\max_p \sum_{e \in E_w^*: p_e = p} d_e$ | Groups workers; comparisons then use all their retained events |
| Self-loop share | Equal adjacent activity labels divided by all adjacent pairs | Within video cases only; all logs |

Table 15 defines the measures and their denominators. For category analysis, the process lists have 113 factory-specific entries and 108 distinct strings across factories. Of the 113 entries, 112 are used in the logs. The unused factory **005** label *motor assembly finishing* has no category assignment in the original map. Every process string actually assigned to an event is mapped. Appendix D of the Additional Material lists the map by factory and marks the unused entry explicitly.

The categories are value-adding production work (VA), material handling and staging (MH), transport (TR), quality inspection (QI), cleaning and maintenance (CM), rework (RW), and documentation (DOC). VA includes packaging and other completion work under the authors' scheme; it is not a direct measurement of customer value. Defect patching is assigned RW because its label and descriptions concern filling or repairing surface defects. This does not establish that it is avoidable or that the upstream process failed a specified quality standard. Supporting work can be necessary even when assigned outside VA.

Runs collapse consecutive same-category rows *within each video*. Their duration is the sum of member durations, which can bridge unrepresented gaps and count overlaps. Run lists are then concatenated across a worker's clips before VA-to-nonVA transitions are counted. This can count transitions across a clip join; the release contains 15 such transitions among eligible workers. These rates are therefore representation-dependent transition measures, not verified interruptions of continuous work. The original script rounds each worker's VA share, transition rate, and mean run duration to one decimal before taking factory medians. We retain this rounding rule. No eligible worker has an empty VA-run list in this release; the original implementation would report zero for that case.

All per-video logs are nonempty. Negative and zero durations are checked separately; no such rows occur. Overlaps remain in time composition and are merged only for coverage. Ambiguous transcript timestamps remain literal. This preserves the original calculations while making their limits visible.

### *6.3. Existing semantic audit protocol*

The archive records 42 sampled events and 42 sampled transcript entries, seven of each per factory. One author made the judgments. The audit record documents a random seed of 42 and factory stratification, but the complete sampling implementation is unavailable. The sampling order, full sampling frame, and handling of timestamp formats cannot be

**Table 16.** Corpus and original transcript diagnostics. Activity and process columns show permitted/used labels. Coverage uses unions divided by parsed endpoint spans, not verified video duration. Alignment is boundary agreement within five seconds under the original parser.

| Factory | Workers | Videos | Events | Activities | Processes | Coverage | Alignment |
|---|---|---|---|---|---|---|---|
| **001** | 11 | 97 | 2,288 | 131/125 | 13/13 | 94.8% | 93.1% |
| **002** | 10 | 54 | 1,541 | 75/67 | 24/24 | 28.2% | 71.8% |
| **003** | 11 | 72 | 2,449 | 55/52 | 12/12 | 93.0% | 86.8% |
| **004** | 10 | 51 | 1,893 | 71/65 | 22/22 | 92.1% | 90.4% |
| **005** | 11 | 22 | 1,103 | 102/97 | 18/17 | 25.0% | 75.0% |
| **006** | 6 | 26 | 1,424 | 61/59 | 24/24 | 14.1% | 62.1% |
| All | 59 | 322 | 10,698 | | | 52.8% | 82.1% |

reconstructed. We report the existing sample as documented and do not claim that it can be redrawn exactly.

For the event sample, the reference is the generated text overlapping the event interval. Code C means fully supported: the label describes the transcribed action in that interval. P means partially supported: the label is plausible but the overlapping text alone is insufficient. X means contradicted. Strict support is $36/42 = 0.857$. The six P cases are retained in the denominator, not treated as fully correct.

For the transcript-entry sample, COV denotes overlap with at least one event, EXC an uncovered entry judged non-operational, and MISS an uncovered entry judged operational. The archive reports $34/(34+4) = 0.895$ after excluding four EXC entries. We call this the *reported entry-overlap rate*, not action recall. One COV entry describes camera tilt covered by a longer event. Thus, the denominator does not establish 38 independently verified operational entries, and overlap does not show that all actions or labels were extracted correctly.

The audit measures consistency with transcripts. It cannot identify visual errors, missed actions, or mistaken object interpretations already present in the transcripts. There is no independent evaluator, inter-rater agreement estimate, or source-video audit. The sample proportions do not bound error in downstream time shares, category assignments, observation selection, or causal interpretation. Appendix E of the Additional Material reproduces all judgments and their notes, with the unverified truncation explanation clearly marked.

## 7. Results and Factory Case Studies

### 7.1. ***RQ1***: *Construction and structural properties*

Table 16 reports the released corpus. All 10,698 events have the required values, nonnegative ordered endpoints, and labels in the relevant factory lists. Start times are nondecreasing within each log. There are no empty logs or zero-duration events. However, 35 adjacent event pairs overlap, 20 in factory **001** and 15 in factory **002**. Passing schema and vocabulary checks therefore does not imply a non-overlapping segmentation. Activity vocabulary use ranges from 89.3% to 96.7%.

### 7.2. ***RQ2***: *Consistency with generated transcripts*

The original parser finds 82.1% of event boundaries within five seconds of a recognized transcript endpoint. It yields an overall coverage diagnostic of 52.8%. Neither statistic is video accuracy. The coverage denominator is heavily affected by the literal multi-hour endpoints listed in Table 12. Alignment can also be low when formats change or when S4 splits a narrated interval. These diagnostics alone do not establish extraction truncation.

**Table 17.** Author judgments of event support and transcript-entry overlap against generated transcripts.

| Sample outcome | Count | Interpretation |
|---|---|---|
| Event fully supported (C) | 36 | Strict transcript support: 36/42, or 85.7% |
| Event partially supported (P) | 6 | Context is plausible; overlapping description is insufficient |
| Event contradicted (X) | 0 | No contradiction among the sampled events |
| Transcript entry overlaps (COV) | 34 | At least one event overlaps; action completeness not established |
| Uncovered, non-operational (EXC) | 4 | Excluded from the archive's reported overlap denominator |
| Uncovered, operational (MISS) | 4 | Reported entry-overlap rate: 34/38, or 89.5% |

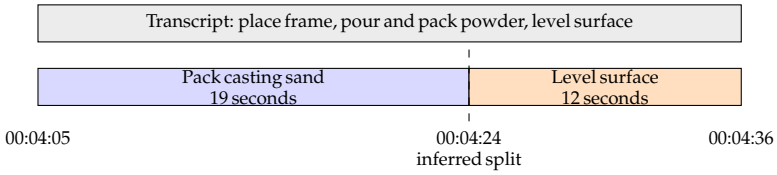



**Figure 5.** Exact transcript and event intervals for the foundry example. The original transcript is line 16 of its text file; the events are data rows 17 and 18 of the corresponding CSV. Bar lengths show recorded durations, not independently checked action boundaries.

Table 17 summarizes the archived judgments. Partial-support examples show the difficulty of checking events against compressed text. Event sample 5 assigns a particular bagging activity where the transcript only states that repetitive assembly and packaging continue. Sample 40 assigns guiding an upper mold half where the text only says that the worker helps with a mold. Recall sample 10 uses a transcript interval from 11:51:00 to 12:59:00 in an anomalous file whose event log ends at 00:20:00. Its MISS judgment is preserved, but the original explanation as long-recording truncation is unverified. These cases show why a single accuracy label would obscure different sources of uncertainty.

### *7.3. An end-to-end foundry example*

Consider factory **006**, worker 003, recording 00003. The S1 transcript interval from 00:04:05 to 00:04:36 describes placing an empty frame on a pattern plate, pouring and hand-packing powder, and using a wooden tool to level the surface. The S2 summary describes preparation, filling, packing, leveling, and compression of molds. It compresses many recordings into a worker-level account. The S3 report supplies process *Sand mold creation*, activity *level powder surface with wooden scraper*, and a compact summary of this worker's molding work.

S4 produces two events within that 31-second transcript interval: *pack casting sand into mold frame*, from 00:04:05 to 00:04:24, and *level powder surface with wooden scraper*, from 00:04:24 to 00:04:36. Both have process *Sand mold creation*. Figure 5 shows what is retained and added. The internal boundary at 00:04:24 is inferred; S1 does not timestamp that transition separately. The events compress placement, pouring, and joint hand-packing into the first activity and add controlled names and a process context.

```
{"events": [{
  "estimated_start_time": "00:04:24",
  "estimated_end_time": "00:04:36",
  "activity": "level powder surface with wooden scraper",
  "process": "Sand mold creation"
}]}
```

Listing 2: Illustrative JSON reconstructed from CSV data row 18. It is not an archived model response.

The linked transcript, worker summary, factory report, and event log preserve the successive representations of this recording. S5 retains the recording identifier and local offsets alongside shifted timestamps. Audit event 41 judges the leveling event fully sup-

**Table 18.** Reproduced time analysis on 320 logs. Left: percentage of summed event duration. Right: factory medians over eligible workers, after the original worker-level rounding. A run sums consecutive same-category rows within a clip; transition counting can cross clips.

| | Process-based category shares (%) | | | | | | | Worker summaries | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Factory | VA | MH | TR | QI | CM | RW | DOC | VA (%) | Trans./h | Mean run (min) |
| **001** | 93.1 | 4.1 | 1.8 | 0.0 | 0.2 | 0.0 | 0.9 | 94.5 | 9.2 | 4.6 |
| **002** | 96.3 | 0.1 | 2.3 | 0.2 | 0.0 | 0.0 | 1.1 | 97.3 | 3.9 | 8.8 |
| **003** | 82.1 | 0.4 | 0.0 | 0.0 | 6.3 | 10.8 | 0.4 | 86.4 | 3.6 | 9.0 |
| **004** | 89.9 | 3.0 | 3.5 | 2.0 | 0.9 | 0.0 | 0.7 | 91.3 | 10.6 | 3.6 |
| **005** | 80.6 | 5.4 | 9.7 | 4.3 | 0.0 | 0.0 | 0.0 | 85.9 | 5.6 | 6.2 |
| **006** | 74.2 | 12.1 | 7.8 | 0.0 | 5.7 | 0.0 | 0.3 | 87.0 | 14.4 | 3.1 |

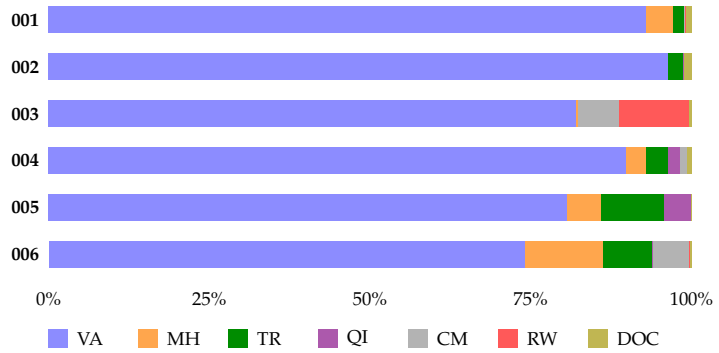



**Figure 6.** Category composition calculated from retained event rows and the released process map. Segments use unrounded duration totals; Table 18 reports rounded percentages. Factories have different task mixes and label schemes, so the plot does not rank efficiency.

ported by its transcript. This supports the label's textual basis, not the exact internal split or a video-based duration measurement.

### 7.4. ***RQ3**: Descriptive analytical capabilities*

#### 7.4.1. Time composition and category transitions

Table 18 reproduces the existing category shares using the actual two-log exclusion rule. The non-VA category share ranges from 3.7% to 25.8% of retained summed event duration. This is neither total worker time nor recoverable capacity. Factory **003** assigns 10.8% to defect patching. Its factory report already describes defect patching in the opening overview and vocabulary. The duration share quantifies this recorded behavior under an explicit category rule.

Figure 6 displays the category shares. The mean-run column is the median across workers of their *mean* VA-run durations. A value of 3.1 minutes in factory **006** is a central tendency; it says nothing about the longest VA episode. Factory **001** has 449 MH/TR rows, a mean row duration of 14.54 seconds, and 14.6 such rows per summed event-hour. Factory **006** has 192 MH/TR rows, a mean of 25.14 seconds, and 28.4 rows per summed event-hour. These are row-based descriptions. Segmentation, repeated-action compression, and clip joins influence their values.

#### 7.4.2. Process flow and discovery

Video-as-case traces describe observation windows. Among the 97, 54, 72, 51, 22, and 26 videos in factories **001** to **006**, there are respectively 97, 53, 70, 51, 22, and 26 distinct activity traces. Self-loops account for 4.3% to 16.8% of within-video directly-follows pairs. These statistics are reproduced directly from the stored labels. A self-loop can reflect separately extracted repetitions or segmentation choices; it is not a count of manufactured items.

**Table 19.** Selected shared-dominant-process groups, using the original eligibility and rounding rules. Ranges summarize worker profiles; they are not estimates of individual efficiency.

| Factory | Dominant process | Workers | VA share | Mean VA run |
|---|---|---|---|---|
| **001** | Manual mechanical assembly | 6 | 89.7 to 97.8% | 1.7 to 6.7 min |
| **001** | Metal stamping | 3 | 90.6 to 98.5% | 3.2 to 7.9 min |
| **003** | Surface finishing and polishing | 5 | 67.3 to 96.9% | 3.3 to 13.7 min |
| **004** | Garment ironing | 2 | 52.8 to 77.1% | 3.5 to 4.1 min |
| **004** | Overlock seaming | 4 | 87.8 to 96.5% | 1.5 to 10.3 min |
| **005** | Manual lathe machining | 2 | 95.0 to 95.8% | 6.9 to 9.4 min |

The released discovery script uses Inductive Miner [32] with noise threshold 0.2, video cases, and activity labels. Appendix C of the Additional Material records the repeated calculation and current dependency version. The recalculated fitness ranges from 0.9544 to 0.9910 and precision from 0.0603 to 0.2679. Fitness measures how the discovered model can replay the inferred traces; precision measures how much additional behavior it permits relative to that log. Both are properties of a model-log pair, not accuracy against actual factory behavior.

Many distinct traces and low precision can arise from flexible work, recording boundaries, abstraction choices, label variation, or extraction mistakes. The present evidence does not separate these causes. No process-level discovery result is reported because its computation and saved output are unavailable. Replacing activity labels with process labels would change the represented behavior and evaluation target. A higher precision after that change would not by itself be a controlled demonstration of better extraction or modeling.

#### 7.4.3. Worker profiles and shared dominant processes

A shared vocabulary permits descriptive comparison within a factory. Table 19 groups workers by their largest process-duration total, then compares VA shares over *all* retained events for those workers. Eligibility requires at least 1,800 seconds of summed event duration. It does not require verified footage duration or equivalent tasks. There are 11, 10, 10, 10, 5, and 5 eligible workers in factories **001** to **006**.

The 29.6 percentage-point range in the surface-finishing group describes different observed profiles. Differences may reflect task mix, product requirements, recording selection, category assignment, or working conditions. Likewise, the small range for two lathe workers cannot rule out differences in behavior. Neither result supports individual performance rankings.

### *7.5. Six descriptive factory cases*

Each case below states its recorded scope, quantitative result, relationship to the report, and unresolved interpretation. Factory descriptions are derived from the reports. Cross-factory rankings are limited by different vocabularies, products, and observation windows.

Factory **001**: hardware stamping and assembly.

The archive has 97 clips from 11 workers; the time analysis retains 95 clips. Manual mechanical assembly accounts for 55.7% of retained summed duration, metal stamping for 18.2%, and component bagging for 11.3%. The report already describes material replenishment, floor storage, and operator transport. The 449 MH/TR rows and their short mean duration quantify the assigned category pattern. They motivate examining material presentation, but do not establish that dedicated handlers, shorter travel, or a different

layout would improve output. Product mix and compressed repetitions remain competing explanations.

Factory **002**: garment finishing and packaging.

There are 54 clips from ten workers, all retained by the event-end rule. Fabric folding, garment bagging, and ironing contribute 18.8%, 17.1%, and 13.2% of summed duration. VA is 96.3%. The report already describes transport, floor staging, and varied folding methods. Quantification shows a small separately categorized handling and transport share, not proof that those behaviors are rare within other processes. Two anomalous transcript endpoints and mixed formats weaken temporal interpretation. Product-specific finishing requirements and category boundaries need examination before any claim of superior efficiency.

Factory **003**: surface preparation and assembly.

The archive covers 72 clips from 11 workers; all clips are retained. Surface finishing and polishing contributes 42.1%, wax pattern assembly 19.2%, and defect patching 10.8%. The report already identifies patching and material transport. The log quantifies patching's share without establishing defect rates, avoidable rework, or financial loss. It also records 101 transport events totaling 1,230 seconds even though the process-based TR share is zero. This case demonstrates how aggregation can hide an extracted behavior. Material types and quality criteria remain uncertain in the report itself.

Factory **004**: garment sewing and finishing.

There are 51 clips from ten workers, all retained. Overlock seaming contributes 23.5%, garment folding 11.0%, and thread trimming 10.8%. The median worker VA-transition rate is 10.6 per summed event-hour. The report already describes tool switching, material handling, and inspection. The logs add a reproducible category pattern and a 24.3 percentage-point VA-share range in the two-worker ironing group. Product specifications, additional finishing duties, and recording windows could explain this difference. The data do not identify which changes to tools or stations would help.

Factory **005**: machining and motor assembly.

The archive contains 22 clips from 11 workers, with only five workers meeting the time threshold. All clips are retained. Manual lathe machining contributes 32.1%, drill press machining 19.1%, and material transport and logistics 9.7%. The report describes staging, fixturing, inspection, and coupled workstations. The log adds duration shares, including 4.3% assigned to dimensional inspection. The two lathe profiles have similar VA shares, but their task equivalence is unknown. Inspection can be necessary, and one transcript has an unresolved endpoint anomaly. The recorded shares cannot measure excess quality work or recoverable machine capacity.

Factory **006**: molding and foundry work.

There are 26 clips from six workers, all retained; five workers meet the time threshold. Sand mold creation, pattern cleaning and coating, and compression molding contribute 17.4%, 14.0%, and 12.2%. The report already describes floor staging, material movement, cleaning, and mold-track operations. The logs assign 19.8% of summed duration to MH/TR and yield a median worker mean VA run of 3.1 minutes. These quantities support a closer study of handling requirements, not a diagnosis of layout as their cause. Necessary mold preparation, differing materials, and segmentation choices are alternatives. Two transcript endpoints require resolution against source metadata.

**Table 20.** Evidence-linked hypotheses. These are candidates for investigation; intervention effects were not measured.

| Observation | Plausible explanation | Competing explanation | Evidence needed |
|---|---|---|---|
| **001**: 449 short MH/TR rows; 5.9% of summed duration | Material presentation creates frequent replenishment | Segmentation splits one handling episode into several rows; task mix differs | Video checks, supply distances, batch quantities, station observations |
| **006**: 19.8% MH/TR; frequent category transitions | Layout or replenishment arrangements require repeated movement | Heavy or fragile materials require necessary support work | Routes, loads, safety requirements, matched task observations |
| **003**: 10.8% defect patching | Upstream defects require corrective work | Patching is a planned finishing requirement | Product specifications, defect criteria, item counts, upstream records |
| **003** and **004**: VA shares differ within dominant-process groups | Conditions or work allocation differ | Product mix, selection, or extraction differences | Comparable products and observation windows; independent video review |
| **005**: two similar lathe VA shares | Common station requirements shape profiles | Similar aggregate shares hide different tasks or behavior | Task-level comparisons and production context |

**Table 21.** Possible meanings of an apparent zero, ordered along the evidence path. Only the last mechanism is directly demonstrated by the factory **003** example; the others are possible mechanisms implied by the pipeline.

| Mechanism | Where information is lost | Needed check |
|---|---|---|
| Behavior absent from the recorded window | Observation selection | Define the window and verify that it is suitable for the question |
| Behavior present but not narrated | Video interpretation or narration | Independent review of source footage |
| Behavior narrated but lost in summaries | S2 or S3 compression | Compare transcripts with worker and factory summaries |
| No suitable vocabulary label | Vocabulary induction | Inspect lists against the behavior of interest |
| Suitable label exists but is not assigned | Event extraction | Compare relevant transcript spans and extracted labels |
| Event exists but category aggregation hides it | Analytical mapping | Compare activity-level rows with process-derived categories |

### *7.6. From observations to operational hypotheses*

Table 20 connects observations to explanations and the evidence needed to distinguish them. Reports identify phenomena; logs can quantify them under a chosen representation. Explaining their causes requires additional evidence. Transport, inspection, cleaning, and documentation are not automatically waste, and the time assigned to them is not automatically available for production. No intervention effect or cost saving is demonstrated by these descriptive results.

## 8. Measurement and Analytical Implications

### *8.1. Observability and the interpretation of zeros*

A zero in an analytical table has several possible causes (Table 21). These causes occur at different stages and require different checks. Requiring a label for transport or waiting improves representability, but does not establish reliable detection or a measured absence. A model may still omit the action, assign another label, or absorb it into a longer episode.

In factory **003**, both *transport loaded tray to storage rack* and *transport bulk material via wheelbarrow* are permitted activity labels and occur in the logs. Together they account for 101 events and 1,230 seconds. No process label is mapped to TR in that factory. Their durations therefore inherit categories such as VA from their assigned processes. The TR category's zero describes the map's output, not the absence of transport or the inability of the activity vocabulary to express it.

To interpret an observed zero as evidence of absence within a window, an analyst would need a defined observation scope, a suitable representation, adequate detection, correct assignment, and an aggregation rule that preserves the behavior. None follows

**Table 22.** Transformation, possible imperfection, analytical consequence, and existing evidence. A prompt or a permitted operation establishes a possible mechanism, not its error frequency.

| Transformation | Possible imperfection | Downstream consequence | Existing evidence |
|---|---|---|---|
| S1: narration | Omitted actions or compressed repetitions; inaccurate times | Row frequency differs from cycle frequency; duration can be distorted | Repetition rule in prompt; seven endpoint anomalies in Table 12; visual omissions unmeasured |
| S2 and S3: summaries and classes | Rare details lost; concepts merged or separated | Vocabulary may omit a needed distinction | Compression requested by prompts; lists available for inspection; information loss not quantified |
| S4: label assignment | Episode forced into the closest permitted label | Membership passes despite weak textual support | Six partially supported audit events, including the bagging example; mechanism not isolated by the audit |
| S4: interval assignment | Internal boundary added without a timed transition in the text | Duration attributed to each action depends on the split | Foundry split at 00:04:24; 35 adjacent overlaps in the full archive |
| S5: consolidation | Filename joins treated as continuous behavior | Added transitions and synthetic elapsed time | Merge code and preserved local offsets; real inter-clip gaps unknown |
| Analysis: category mapping | Recorded activity hidden by its process category | Apparent zero for a behavior present in the log | Factory **003** has 101 transport events but no process mapped to TR |

from label membership alone. Conversely, an observed event can establish what the representation contains while leaving the event's visual accuracy unmeasured.

### *8.2. Uncertainty propagation and limits of provenance*

Errors can propagate along both branches of Figure 3. A mistaken object description can shape the worker summary, the factory vocabulary, and the final event label. Agreement among those outputs can therefore reflect a shared error. Quality-aware construction [9] and event-log imperfection patterns [23] establish the broader concern. Table 22 shows how it arises in this construction, separating observed examples from possible mechanisms.

These mechanisms occur within an observation scope that is itself incomplete: the selection protocol and source durations are unavailable. No stage-wise error rates are calibrated, and structural checks cannot quantify visual errors. Worker and factory differences may therefore reflect observation selection as well as representational choices.

Readable text is useful evidence, but it is weaker than structured event provenance. Existing records identify a source file and local interval. They do not store supporting transcript-span identifiers, alternative labels, calibrated confidence, or a flag that distinguishes an inferred internal boundary from a copied transcript endpoint. Finding the source of an event therefore requires reading and matching intervals. The foundry example can be traced manually, yet its CSV does not record why the split at 00:04:24 was chosen.

A future schema could add transcript-entry references, a boundary-origin field, and vocabulary and prompt versions. Data-quality annotations could retain known issues for downstream analysis [24]. Explicit alternative labels or bounded times would instead extend the event representation toward uncertain data [25,33]. Such alternatives need evidence: the five-second matching tolerance does not imply that an event boundary lies within five seconds of its true value. Calibrated confidence values are unavailable for the existing events. Provenance can explain how a conclusion was reached; it does not certify that every preceding observation was correct.

### *8.3. Missing time does not imply a one-sided bound*

The observed share outside VA is sometimes described as a lower bound because waiting and other supporting work can be omitted. That interpretation also assumes that omitted time is sufficiently non-VA. Missing VA time can instead raise the observed non-VA share.

Proposition.

Consider an idealized observation with $C > 0$ seconds of correctly classified, non-overlapping observed time, of which $N$ seconds are non-VA. Let $U \geq 0$ seconds be unobserved. If observed and unobserved time partition a known target window, then its full-time non-VA share satisfies

$$\frac{N}{C+U} \leq p_{\text{nonVA}} \leq \frac{N+U}{C+U}. \tag{4}$$

*Proof.* Let $x$ be the non-VA duration in the unobserved part. Since $0 \leq x \leq U$, the full share is $(N+x)/(C+U)$ and has the stated extrema. Both are attainable: all missing time may be VA or all may be non-VA.

For $U > 0$, the observed ratio $N/C$ is a lower bound only if $x/U \geq N/C$. Thus, merely knowing that some waiting is missing is insufficient. The interval width in Equation 4 is $U/(C+U)$, which shows how unknown coverage limits a share even with perfect observed labels.

This is a conceptual result, not an empirical confidence interval for this dataset. Source durations and therefore $U$ are unavailable; labels are inferred; and summed event durations include overlaps. Substituting the coverage diagnostic or the 97.1 retained summed event-hours into this proposition would violate its assumptions. The result identifies the additional evidence needed before making a bound claim.

### *8.4. Why aggregation changes the question*

A duration-weighted factory share estimates the composition of the retained event rows. A median worker share gives equal weight to each eligible worker. They answer different questions. In factory **006**, for example, the factory VA share is 74.2%, while the median worker VA share is 87.0%. Different observation amounts and work profiles can produce this difference without either statistic being erroneous.

Likewise, process-based and activity-based categories need not agree. Counting rows emphasizes the extraction granularity; counting runs suppresses adjacent labels; measuring interval unions removes overlap but does not resolve classification conflicts. None is a universal correction. A defensible analysis states its unit, denominator, case rule, handling of gaps, and interpretation before drawing operational conclusions. The released measurements serve as a reproducible example of those choices and their limits.

## 9. Discussion and Conclusions

### *9.1. Main findings*

Egocentric recordings can be organized into a shared representation of manual work through narration, factory vocabulary induction, and interval-event extraction. For **RQ1**, the released logs satisfy the reported field, ordering, and vocabulary checks. For **RQ2**, the author audit finds substantial agreement with generated transcripts, while accuracy against the videos remains unmeasured. These results distinguish structural consistency and textual support from correct observation.

For **RQ3**, the logs support descriptions of work composition, local sequences, and worker profiles. They quantify behaviors described in the reports and identify questions for further investigation. For **RQ4**, the examples show how representation choices affect these measurements. In factory **003**, recorded transport disappears from the transport category because of the process-based map. In the foundry example, an inferred boundary divides a narrated episode between activities. Synthetic clip joins can add transitions, and missing

time can move a category share in either direction. Interpreting a result therefore requires its category rule, event boundaries, case definition, and denominator.

### *9.2. Limits and responsible use*

The vocabulary is induced from the same factory corpus used for assessment, so the results do not establish transfer to unseen factories. The single-author audit uses generated text and cannot reveal visual errors shared by the transcripts and logs. Missing source durations, selection records, and historical execution details limit temporal interpretation, representativeness, and exact regeneration. The released artifacts still permit reproduction of the descriptive calculations.

Different task mixes, observation windows, and category assignments can explain differences between worker profiles. Product identifiers and production outcomes are absent, and no intervention was evaluated. The results consequently support hypotheses about process design, not causal diagnoses, efficiency rankings, or estimates of recoverable capacity.

Worker identifiers are pseudonymous and may still enable identification when combined with local information [34]. Use should remain tied to an agreed analytical purpose, with suitable aggregation and review by the people whose work is represented. These inferred profiles do not support employee appraisal, pay, or disciplinary decisions. The source card does not document the original consent procedure; this secondary study collected no new consent.

### *9.3. Next steps*

The linked narrations, summaries, vocabularies, and logs provide a basis for testing alternative event representations. Priorities are independent video assessment with multiple evaluators, preservation of source durations, and explicit links from events to supporting transcript spans. Product references and real calendar times would be needed to extend the resource to synchronized production-flow analysis.

## Additional Material

The accompanying Additional Material contains the evidence and assessment scope (Appendix A), reproducibility information and configuration (Appendix B), activity-level discovery results (Appendix C), complete operational-category map (Appendix D), all archived audit judgments (Appendix E), and full extraction prompts (Appendix F).

## Declarations

**Author Contributions:** Alessandro Berti carried out the experiments. Wil M. P. van der Aalst reviewed the manuscript and suggested improvements.

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**Institutional Review Board Statement:** This study is a secondary analysis of the existing Egocentric-10K dataset. It involved no recruitment of participants, intervention, or new video collection. No ethics approval or exemption record for this secondary analysis is reported.

**Informed Consent Statement:** No participants were recruited or contacted for this secondary analysis. The source videos were obtained from the existing Egocentric-10K dataset. The authors did not collect new consent, and the source dataset card does not report the original consent procedure.

**Data Availability Statement:** The pipeline, prompts, transcripts, factory vocabularies, derived event logs, and assessment code are available at https://github.com/fit-alessandro-berti/annotated-egocentric-10k-dataset. The source videos are available from Build AI's Egocentric-10K dataset at https://huggingface.co/datasets/builddotai/Egocentric-10K, under its access terms and Apache License 2.0. The repository contains derived artifacts and does not redistribute the source videos.

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**Conflicts of Interest:** Wil M. P. van der Aalst is affiliated with Celonis, a process mining software company. No other conflicts of interest are reported.

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